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CS & IT

Computer Networks

IPv4 Addressing

DPP: 1

Q1 In the IPv4 addressing format, the number of networks allowed under Class B addresses are _____.

(A) 2^8 (B) 2^{10}
(C) 2^{14} (D) 2^{16}

Q2 In a classless network, consider broadcast address (special IPv4 address) of the network is "100 . 125 . 159 . 255", which can be correct network address (special IPv4 address) of the network?

(A) 100 . 125 . 96 . 0
(B) 100 . 125 . 120 . 0
(C) 100 . 125 . 128 . 0
(D) 100 . 125 . 148 . 0

Q3 In the network "202 . 101 . 111 . 210 / 27", the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

Q4 Consider network mask of a given network is "255 . 255 . 240 . 0", how many maximum number of hosts can be in the network?

Q5 The network mask for a particular network is "255 . 255 . 240 . 0". Which of the following pairs of IP addresses could belong to this network?

(A) 210.130.148.12 and 210.130.168.30
(B) 210.130.148.14 and 210.130.158.32
(C) 210.130.158.16 and 210.130.168.34
(D) 210.130.164.18 and 210.130.168.38

Q6 An organization has a class B network and wishes to form subnets for 120 departments. The subnet mask would be:

(A) 255.255.240.0 (B) 255.255.248.0
(C) 255.255.254.0 (D) 255.255.255.0

Q7 If a class B network on the Internet has a subnet mask of 255.255.252.0, what is the maximum number of hosts per subnet?

Q8 A subnetted Class C network has the broadcast address : 220.210.210.31, which can be its subnet mask ?

(A) 255.255.255.128
(B) 255.255.255.192
(C) 255.255.255.224
(D) 255.255.255.240

Q9 Consider a class C network address is 203.204.205.0 with four bit subnetting. Identify which can be a valid subnet address?

(A) 203.204.205.8 (B) 203.204.205.16
(C) 203.204.205.24 (D) 203.204.205.32

Q10 Which one of the following CIDR prefixes exactly represents the range of IP addresses 170.120.200.0 to 70.20.207.255 ?

(A) 170.120.200.0/20
(B) 170.120.200.0/21
(C) 170.120.200.0/22
(D) 170.120.200.0/23

Q11 Consider two hosts are connected through four intermediate routers, how many total number of

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network segments in this scenario?

Q12 Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation "/N" in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	Next hop router
120.130.140.0/24	R1
120.130.140.128/25	R2
120.130.140.64/26	R3
120.130.140.192/26	R4

IP address 120.130.140.205 will forward to the router. Identify the next Hop

(A) R1 (B) R2

(C) R3 (D) R4

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Answer Key

Q1	(C)	Q7	1022
Q2	(C)	Q8	(C, D)
Q3	222	Q9	(B, D)
Q4	4094	Q10	(B)
Q5	(B, D)	Q11	5
Q6	(C)	Q12	(D)

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Hints & Solutions

Q1 Text Solution:

Class B IPv6 Network

Net ID = 16 bit

IP Add = 10 . . . . .

Number of networks = 2^{14}

Q2 Text Solution:

Broadcast Add = 100 125 159 255

100 125 100 1 1 1 1 . 1 1 1 1 1 1 1 1

Host ID maximum 13 bit

Network address

100 . 125 . 1 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0

100 . 125 148 . 0

100 . 125 . 10010100 . 0 0 0 0 0 0 0 0

Host ID

Q3 Text Solution:

202 . 101 . 1 1 1 . 210/27

202. 101. 111. 1 1 0 1 0 0 1 0/27

Prefix host ID

202 . 101 . 111 . 110 1 1110/27

202.101.111.222

Q4 Text Solution:

Network = 255.255.240.0

255.255.1111 0000 0000 0000

Net ID (20 bit) Host ID (12 bit)

Size of network = (2^{12} – 2) hosts per netowrk = 4094

Q5 Text Solution:

148 → 10010100

240 → 11110000

144 ← 1001000

168 → 10101000

240 → 11100000

160 ← 10100000

158 → 10011110

240 → 11110000

144 ← 10010000

164 → 10100100

240 → 11110000

160 ← 10100000

Q6 Text Solution:

Class B netowkr : -

Net ID = 16 bits

Subnet ID = $\lceil \log_2 (120) \rceil$ bits

= 7 bits

Subnet mask:

255.255.11111110.00000000

255.255.254.0

Q7 Text Solution:

Class B Network:-

Net ID = 16 bits

Subnet mask = 255.255.252.0

255.255. 111111 00.00000000

Net ID 16 bits subnet ID 6 bit 16 bit host ID

Size of subnet = (2^{10} – 2) hosts per subnet = 1022

Q8 Text Solution:

Class C network, subnet directed boradcast address

= 220.210.210.31 = 220.210.210. 000 111111

Net ID 24 bit

Host Id

As per maximum 5 bit host ID

Subnet mask = 255.255.255.224

As per 4 bit host ID

subnet mask = 255.255.255.240

Q9 Text Solution:

Class C network address = 203.204.205.0

203. 204. 205 - - - - - - - -

Net ID 24 bit Subnet ID 4 bit HostID 4 bit

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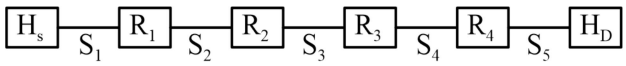
1st subnet → 0000 0000 = 0
2nd subnet → 0001 0000 = 16
3rd subnet → 0010 0000 = 32

Q10 Text Solution:

170.120.11001000.00000000
170.120.11001111.11111111
170.120.11001000.00000000
170.120.200.0/21

Q11 Text Solution:



Q12 Text Solution:

Destination IP address = 120.130.140.205 → R4
120.130.140.205
255.255.255.192
130.130.140.192

205 → 11001101
192 → 11000000
192 ← 11000000

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